Incubator



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Incubator; Incubators

What it is used for

A laboratory incubator maintains a controlled temperature for growing or holding approved biological samples or cultures. Correct temperature settings, suitable loading, and contamination control help maintain sample integrity and reliable experimental conditions.

Main parts

- Insulated chamber provides the controlled environment.
- Heating element raises the chamber temperature.
- Temperature sensor measures the internal temperature.
- Thermostat or controller regulates the set temperature.
- Display shows the setpoint and measured temperature on equipped models.
- Shelves or racks hold sample containers.
- Door and gasket help maintain chamber conditions.
- Fan circulates air in forced-air models.

Before you start

1. Only trained personnel should use the incubator according to the laboratory procedure.
2. Read the manual and confirm the required temperature and operating range for the sample.
3. Inspect the chamber, shelves, door gasket, power cord, and controls for damage.
4. Ensure the chamber is clean and suitable for the intended sample type.
5. Check calibration or temperature verification status and confirm the display is functioning.
6. Prepare clearly labelled, compatible containers and follow the required containment procedure.

How to use

1. Place the incubator on a stable surface with adequate ventilation around it.
2. Connect the unit to the specified power supply and switch it on.
3. Set the approved temperature and allow the chamber to stabilise before loading samples.
4. Arrange labelled containers with adequate space for air circulation and temperature uniformity.
5. Close the door securely and avoid unnecessary opening during incubation.
6. Monitor and record temperature at the required intervals.
7. Check samples according to the approved schedule without disturbing other containers unnecessarily.
8. If a temperature alarm or deviation occurs, follow the laboratory procedure and assess sample validity.

9. After use, remove samples using the required containment practices and switch off the unit if instructed.
10. Clean and dry the chamber as required and record maintenance or temperature deviations.

⚠ Safety

- Never incubate unknown or hazardous samples without approved containment and handling procedures.
- Do not use the incubator if electrical parts, the door, or the temperature controls are damaged.
- Avoid overcrowding the chamber or blocking ventilation openings.
- Use appropriate PPE when handling biological samples and contaminated containers.
- Do not use the incubator for flammable or volatile materials unless specifically designed and approved for them.
- Report temperature deviations, spills, or equipment faults promptly.

Care & storage

- Clean the chamber and shelves with approved materials compatible with the incubator.
- Inspect the door gasket, shelves, sensor, and ventilation openings regularly.
- Keep records of temperature checks, calibration, cleaning, and maintenance.
- Arrange authorised servicing for persistent temperature instability, alarms, or electrical faults.

Common mistakes to avoid

- Loading samples before the chamber reaches the required temperature.
- Setting an incorrect temperature or ignoring temperature deviations.
- Overloading shelves and restricting air circulation.
- Using unlabelled samples or failing to follow contamination control procedures.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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